



The effect on soot and its gas precursors of doping ethylene with 2,2,4,6,6-pentamethyl-heptane in the nitrogen-fuel stream of a laminar non-premixed Planar Mixing Layer Flame (PMLF)

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ABSTRACT

Synthetic Aviation Turbine Fuels (SATFs) are promising for reducing soot emissions from the aviation sector and diversifying Jet Fuel (JF) sources. Accurately predicting the combustion and emissions behavior of SATFs (and other JFs) necessitates robust experimental databases to elucidate the chemistry of long-chain iso-paraffins, which can compose up to two-thirds of SATF blends and whose behavior is considered to be well-represented by that of iso-dodecane isomers. This study characterizes two laminar non-premixed Planar Mixing Layer Flames (PMLFs) with mild soot loads fueled by nitrogen-diluted ethylene, pure and doped with 2,2,4,6,6-pentamethyl-heptane, respectively. The two PMLFs have the same stoichiometric mixture fraction and total hydrocarbon mole fraction in the fuel stream ($X_{F,F} = X_{C_{12}H_{26},F} + X_{C_{12}H_{26},F} = 0.260$), resulting in nearly the same maximum temperature ($T_{max} \approx 1800$ K) and simple identification of the effects of doping. Importantly, any horizontal PMLF cross-section has a self-similar structure that can be modeled as an equivalent One-Dimensional Counterflow Flame (1D-CF) with vanishingly small strain rate (a). The cross-section at a Height Above the Burner (HAB) of 50 mm is characterized in terms of C_0 - C_{18} gas species using capillary sampling followed by GC-MS analyses. Laser-Induced Emission Spectroscopy (LIES) quantifies the soot volume fraction (f_v) profiles at HAB=25 and 50 mm where Elastic Laser Light Scattering (E-LLS) is performed to determine the a of the equivalent 1D-CFs and the profile of the E-LLS equivalent diameter ($d_{6,3}$) of soot. The substitution of 1500 ppm of ethylene with 2,2,4,6,6-pentamethyl-heptane causes an increase of ≈ 1.5 in the concentrations of several polycyclic aromatic hydrocarbons and f_v . Concurrently, the measured $d_{6,3}$ doubles in the oxidizer stream, yet remains the same in the fuel stream, at HAB=50 mm. Instead, at HAB= 25 mm, the iso-dodecane doping does not affect the $d_{6,3}$ profile in either stream. The experimental results partially validate the chemical reactions and soot formation kinetic model developed at Lawrence Livermore National Laboratory and provide directions to further improve its predictions.

1. Introduction

Soot emissions are among the most significant anthropogenic contributors to atmospheric pollutant aerosols [1], and, as a result, the investigation of its formation in flames has been the subject of intense scientific research for several decades [2–10]. Yet the mechanisms of the inception step involving the transition of gaseous precursors into soot nuclei are still an object of scientific debate [5–7,9,10]. In the aviation sector, the emission of pollutant soot [11] within the turbine exhaust

wake also promotes the formation of contrails and cirrus clouds [1]. Since Synthetic Aviation Turbine Fuels (SATFs) can be produced from a variety of feedstocks, the transition from traditional aviation Jet Fuels (JFs) to “drop-in” SATFs is considered a viable solution for diversifying energy sources and, importantly, reducing soot emissions [12,13]. This is because studies in non-premixed Counterflow Flames (CFs) have shown that the sooting tendencies of most SATFs are lower than those of traditional JFs [14,15]. Still, the widespread use of SATFs requires developing detailed kinetic models to predict their combustion

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chemistry and sooting behavior (and those of JFs), via experimental studies [15,16]. These efforts most often focus on surrogate mixtures composed of fewer than 10 pure components, which mimic the combustion characteristics of real, much more complex, fuel blends. Since SATF blendstocks can contain a high percentage of iso-paraffins [14–16], their surrogates often include iso-dodecane isomers (iC12) to mimic the average composition, reactivity, and other relevant bulk properties of the real fuel blends. Thus, the experimental characterization of the complex oxidation and pyrolysis chemistry of highly branched alkanes is needed to validate detailed SATF chemical kinetic models. Recently, Chatterjee et al. [17] performed measurements of soot volume fraction in an iC12-fueled non-premixed CF to validate the detailed kinetic model developed at Lawrence Livermore National Laboratory (LLNL). Additionally, the effects of the fuel molecular structure were studied by Fang et al. [18], who provided a validation database for the autoignition of several highly branched iso-paraffins.

Canonical laminar flame configurations are used for most detailed studies on soot inception, including those from iC12 combustion [19]. Soot formation is investigated predominately in Burner-Stabilized Premixed Flames (BSPF) which have mild spatial gradients and are suitable for intrusive sampling. Also, soot formation in BSPFs can be followed in time by performing measurements at different Heights Above the Burner (HABs) [7,9,10,20,21]. Nonetheless, most combustion applications for which soot emissions are of relevance, including aviation propulsion ones, rely on non-premixed combustion which is typically investigated in co-flow Flame (co-F) [8,22] or CF [4,14,17,23] configurations, neither of which are amenable to intrusive sampling. Other drawbacks of the co-F configuration include the imperfect control of their boundary conditions, the inability to separate pyrolysis and oxidation, and the thermochemical advection time history experienced by the flame products being highly polydisperse and complicatedly affected by buoyancy, calling for complex 2D-CFD modeling to interpret the results [21]. On the other hand, CFs have an easy-to-model 1D structure with well-defined boundary conditions and pyrolysis separated from oxidation but pose strict limits on its sampling and the values of flame strain rate (i.e., global advection times) that can be investigated experimentally [4,24].

The Planar Mixing Layer Flame (PMLF) configuration recently introduced by Ashour and Carbone [25] overcomes most of the co-F and CF limitations thanks to its stabilization based on flow impingement and buoyancy. As a result, the PMLF consists of an unconfined, steady, and relatively thick non-premixed flame with a self-similar boundary layer structure. The PMLF is amenable to intrusive sampling, with pyrolysis separated from oxidation, and advection times of tens of milliseconds, typical of most combustion applications. Importantly, the horizontal cross-sections of a PMLF can be easily modeled using 1D-CF solvers with low strain rates thanks to their self-similarity [25,26].

This work investigates the effects on soot and its precursors of doping the nitrogen-diluted ethylene fuel stream of a PMLF with 1500 ppm of 2,2,4,6,6-pentamethyl-heptane (iC12). In-situ laser diagnostics are implemented to measure the horizontal profiles of the volume fraction and the elastic light scattering equivalent spherical particle diameter of soot, as well as the profile of the elastic light scattering coefficient, determined by temperature and composition fields. Measurements of several gaseous species, including the parent fuels and their combustion products as heavy as Polycyclic Aromatic Hydrocarbons (PAHs) composed of up to 5 rings, are performed via capillary sampling followed by Gas Chromatography - Mass Spectrometry (GC-MS) to resolve the gas-phase chemical structure of the PMLF at HAB = 50 mm. Instead, soot is characterized in the PMLF horizontal cross-sections at two HABs (i.e., = 25 and 50 mm) to study the effect of residence time and flame strain rate. Results in a base, undoped PMLF enable quantifying the effect of the iC12 combustion and sooting chemistry [27,28].

2. Experimental and computational methods

2.1. The base Planar Mixing Layer Flame (PMLF)

Plug flows at 0.20 m/s and 298 K exit adjacent $25 \times 50 \text{ mm}^2$ slot nozzles and are composed of C_2H_4 (mole fraction is $X_{\text{FF}} = 0.260$) and O_2 ($X_{\text{OO}} = 0.179$) diluted in nitrogen as fuel and oxidizer, respectively. As a result, the stoichiometric mixture fraction (Z_{st}) of the stabilized PMLF is 0.183, a choice ensuring that soot formation is decoupled from oxidation [25,29]. A 12 mm thick square annulus surrounds the reactant nozzles, delivering a plug flow of sheath nitrogen at 0.20 m/s and 298 K. The flow impinges at a Height Above the Burner (HAB) of 80 mm onto a $75 \times 75 \text{ mm}^2$ stainless-steel plate equipped with a $50 \times 14 \text{ mm}^2$ slit opening parallel to the splitter between the reactant nozzles, but shifted 2 mm toward the oxidizer one, to exhaust the hot flame products via buoyancy and anchor their position. Fig. 1 shows a digital image of the studied base PMLF on the left and a cross-section of the burner on the right. Further information on the base PMLF can be found in [25]. The digital image shows the blue chemiluminescence which is expected to coincide with the horizontal maxima of the heat release rate [25,30], and the faint orange luminosity typical of mild soot loads to its left toward the fuel nozzle [7]. During the measurements, digital imaging tracks the horizontal distance between the measuring point and the blue layer, as set via a horizontal translation stage attached to the burner.

2.2. PMLF doped with 2,2,4,6,6-pentamethyl-heptane (iC12)

The doped PMLF is stabilized upon replacing 1500 ppm molar of ethylene in the fuel stream with the same amount of pre-vaporized 2,2,4,6,6-pentamethyl-heptane (iC12). As a result, the total fuel mole fraction in the fuel stream, $X_{\text{FF}} = X_{\text{C}_2\text{H}_4,\text{F}} + X_{\text{C}_{12}\text{H}_{26},\text{F}} = 0.260$, is the same as in the base PMLF, but the oxygen mole fraction in the oxidizer stream (X_{OO}) is increased to 0.183 to compensate the extra H and C content of iC12 compared to the C_2H_4 it replaces and keep $Z_{\text{st}} = 0.183$. As in previous studies of our group [27,28], pre-vaporization is achieved with an electrospray stabilized by enhancing the iC12 electrical conductivity with trace amounts of STADIS 450 (<1 ppmv in liquid iC12). The electrospray is operated in cone jet mode and dispersed in a 4 slpm stream of nitrogen preheated to 140 °C. The fine mist stream crosses a downstream 1 m long copper tube kept at 220 °C to facilitate complete droplet evaporation before the flow is mixed in a tee with the remaining gaseous components of the fuel stream. The total fuel stream then crosses another 2 m long copper tube naturally cooled by ambient air and connected to the burner. Gas chromatography verifies that the mass flow of iC12 at the outlet of the burner is the same as that exiting the heating chamber.

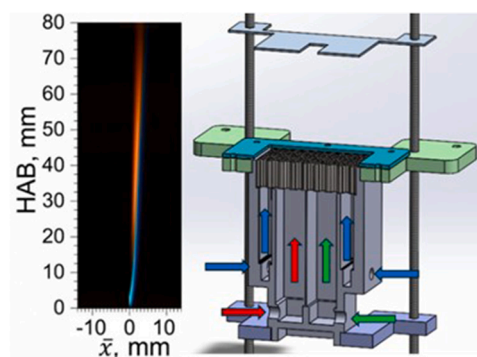


Fig. 1. (Left) Digital image of the PMLF. (Right) Cross-section of the burner with red, green, and blue arrows indicating the fuel, oxidizer, and shielding nitrogen streams, respectively. (For interpretation of the references to color in the legend, the reader is referred to the Web version of this article).

2.3. Optical diagnostics and gas speciation via capillary sampling

A Nikon D750 digital camera with a Nikkor prime lens (i.e., $f = 85$ mm/2.0 during capillary sampling or $f = 24$ mm/2.8 during laser measurements) coupled to an achromatic close-up lens (Sigma AML72–01) is used to image the flame with a line of sight parallel to the splitter of the fuel and oxidizer nozzles. At least 30 images are taken to quantify the uncertainty of the horizontal position of the measuring point with error bars bracketing the $\pm$ standard deviation interval of its horizontal distance from the PMLF blue luminosity in each image [25].

In-situ laser diagnostics [31–36] are implemented by illuminating the measuring point with the vertically polarized 3rd harmonic ($\lambda = 355$ nm) of an Nd:YAG laser (Quantel Ultra 100) crossing the PMLF horizontally (from the oxidizer to the fuel side). The laser beam cross-section at the measuring point is reshaped into a 2 mm tall and 6 mm wide ellipse by two couples of cylindrical lenses. The light emission induced by the laser pulse is collected parallel to the flame luminosity and collimated by a $f = 200$ mm/4 achromat spherical lens before being redirected by a planar mirror and refocused by another $f = 200$ mm/4 achromat spherical lens onto the 100 μ m slit of an Andor Shamrock 163 spectrograph coupled to an Andor iStar sCMOS gated intensified camera.

Laser-Induced Emission Spectra (LIES) [33–36], mostly due to incandescence and, possibly, broadband fluorescence [26,37], are measured in the 500–650 nm spectral interval with 150 ns camera gating time. The measured LIES are converted to soot volume fraction (f_v) via calibrations performed in an extensively characterized BSPF [20] based on the similarity of the spectra measured in both flames at locations with comparable f_v shown in the supplementary material. Error bars bracketing the $\pm$ standard deviation interval of the results are obtained by investigating three different spectral sub-intervals [26,36]. Also, the Elastic Laser Light Scattering (E-LLS) coefficient of the flame products ($Q_{0,0^{\circ},f}$) [31,32,36] is measured with 12 ns camera gating time, upon installing a thin film polarizer between the twin spherical lenses to collect only the scattered light with the same polarization as that of the laser, and by converting the camera counts using calibrations performed in pure gases [31,36]. The Rayleigh scattering coefficient of the gas mixture ($Q_{0,0^{\circ},g}$) can be calculated knowing the cross-sections and the number concentration [7,17] of each species, as listed in the literature [36,38] and estimated by computational modeling, respectively. The value of the 2,2,4,6,6-pentamethyl-heptane E-LLS cross-section has been measured directly upon pre-vaporizing and mixing it in an ultrapure He stream. The measured $Q_{0,0^{\circ},f}$ can be converted to the Rayleigh scattering equivalent diameter ($d_{6,3}$), according to Eq. (1) [31,32,36] upon assuming the scattering function of soot to be $F(m) = 0.20 \pm 0.08$ according to a survey of values of the soot complex refractive index (m) reported in the literature [39–42].

$$d_{6,3} = \left[\frac{Q_{0,0^{\circ},f} - Q_{0,0^{\circ},g}}{\frac{6\pi^3}{4\lambda^4} f_v F(m)} \right]^{1/3} \quad (1)$$

Finally, the chemical gas structure of the PMLF horizontal cross section at HAB = 50 mm is probed using fused silica capillary sampling of its gas products (ID/OD=0.196/0.354 mm) followed by their GC-MS (Agilent 5977B) analyses [7,25]. The samples are drawn through a heated sampling line at 473 K and stored (less than 30 s) at 300 torr before being injected into the GC-MS system which is customized to quantify stable hydrocarbons at least as heavy as 276 Da and more abundant than 5 ppbv [25]. As a result, more than forty-five species are accurately quantified to investigate the chemistry of the pyrolysis of iC12 in the studied doped PMLF. For brevity, results are presented herein only for 29 of them.

2.4. Computational modeling

Each horizontal cross-section of the PMLF can be modeled as an equivalent One-Dimensional CF (1D-CF) [25,26]. This provides a computationally effective method for validating large chemical kinetic and soot formation models which are hard to test in 2D-CFD. Herein, the planar 1D-CF solver of the Chemkin Pro (Ansys LLC) package has been coupled to the detailed chemical kinetic model developed by LLNL [17] to test the latter against the PMLF experimental results. The solver accounts for mixture averaged and Soret diffusion as well as optically thin gas radiation losses [30]. Equal plug flow velocities are assigned to the fuel and oxidizer inlets ($V_F = V_O = a \cdot L / 2$). The velocity is related to the strain rate (a) of the equivalent 1D-CF, with an arbitrarily large nozzle separation distance (i.e., $L = 80$ mm), which is equivalent to the PMLF horizontal cross-section [25] according to two criteria. First, a must be selected such that the width of the temperature profile of the 1D-CF is equal to that of the PMLF cross-section. Second, the fuel and oxidizer mole fractions of the 1D-CF must be perturbed while keeping Z_{st} constant to make its maximum temperature (T_{max}) the same as that of the PMLF horizontal cross-section [25]. Both criteria are satisfied by matching the profile of the light scattering coefficient ($Q_{0,0^{\circ},g}$) of the gas species predicted by the 1D-CF to that measured in the PMLF cross-section at the desired HAB [26], as listed in Table 1 and detailed in the results section.

The LLNL kinetic model [17] has been developed to mimic soot formation in CFs fueled by iC12 and considers 12,366 reactions between 1,386 species. The classes of reactions involving soot include those mimicking its inception, surface growth, coagulation, and oxidation, as detailed in published work [17]. Soot is modeled using the discrete sectional method with 20 sections of lumped stable and radical species denoted as BIN_n and $BINJ_n$, respectively. The subscript n sorts the BIN s according to the number of carbon atoms, from 20 up to 10^7 Da. Heavy PAHs and their clusters are considered with n up to 4, the soot nuclei are the BIN s with n larger than 5 and up to 12, and aggregates are accounted for with even larger n , up to 20. According to such classification, the soot volume fraction (f_v) is equal to that of $BIN_{n,s}$ and $BINJ_{n,s}$ with n from 5 to 20, which can be computed based on the mass balance in Eq. (2).

$$f_v = \frac{\sum_{n=5}^{20} (MW_{BIN_n} \cdot X_{BIN_n} + MW_{BINJ_n} \cdot X_{BINJ_n})}{\rho_{soot}} \frac{\rho_{gas}}{MW_{gas}} \quad (2)$$

In Eq. (2), the gas density (ρ_{gas}), average molecular weight (MW_{gas}), and mole fractions of BIN_n (X_{BIN_n}) and $BINJ_n$ (X_{BINJ_n}) are extracted from the output file of the 1D-CF model. The molecular weights of BIN s (MW_{BIN_n}) and $BINJ$ s (MW_{BINJ_n}) are an input of the LLNL model [17], and the value of the soot density is assumed as $\rho_{soot} = 1.45$ g/cm³, which is an average between those typical of incipient and mature soot [5]. A summary of all experimental and 1D-CF boundary conditions is presented in Table 1.

3. Results and discussion

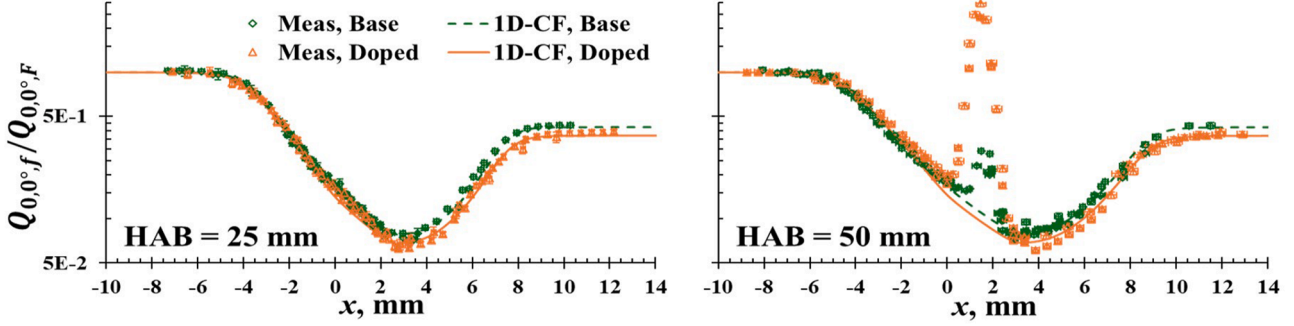
3.1. The PMLF and its equivalent 1D-CFs

Fig. 2 shows the profile of the experimental and modeled E-LLS coefficient ($Q_{0,0^{\circ},f}$) of the flame products scaled over its value for the corresponding fuel stream ($Q_{0,0^{\circ},F}$). The scaling provides a fair comparison since $X_{C2H4,F}$ is slightly smaller in the PMLF cross sections compared to their equivalent 1D-CFs, as listed in Table 1 [25]. In Fig. 2, the lines depict the contribution to $Q_{0,0^{\circ},f}$ due to the gaseous species ($Q_{0,0^{\circ},g}$) predicted by the 1D-CF model whereas the points quantify the $Q_{0,0^{\circ},f}$ measured in the base and doped PMLFs. The x-axis origin is set at the Gas Mixing Interface (GMI) for the PMLF and the Gas Stagnation Plane (GSP) for the 1D-CFs, since they are the virtual barriers between the gas molecules in the fuel and oxidizer streams that molecules can cross only via diffusion [25]. The width of the $Q_{0,0^{\circ},f}$ profile increases as

Table 1

Boundary conditions of the PMLF and 1D-CFs equivalent to its cross-sections at HAB = 50 and 25 mm.

Base-Doped	$X_{C_2H_4,F}$	$X_{C_{12}H_{26},F}$	$X_{C_2H_6,F}$ Impurity	$X_{O_2,O}$	V_O , m/s	V_F , m/s	1D-CF a , s ⁻¹	1D-CF L , mm	1D-CF T_{max} K
Experimental PMLF	0.25995-0.25845	0-0.00150	0.00005	0.17909-0.18313	0.200	0.200	–	–	1793 [25] -N/A
1D-CF, HAB=50mm	0.27145-0.26988	0-0.00157	0.00005	0.18721-0.19139	0.232	0.232	5.8	80	1781-1796
1D-CF, HAB=25mm	0.27085-0.26929	0-0.00156	0.00005	0.18679-0.19095	0.324	0.324	8.1	80	1798-1813

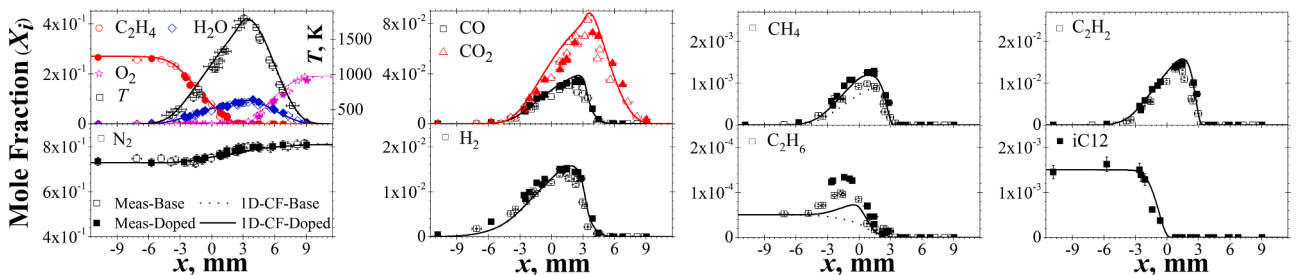
**Fig. 2.** Profile of the E-LLS scattering coefficient in the investigated PMLF horizontal cross-sections, as measured and computed for their equivalent 1D-CF. (For interpretation of the references to color in the legend, the reader is referred to the Web version of this article).

the flame thickness increases at increasing HABs [25], and the strain rate (a) of the 1D-CF is tuned to reproduce the experimental results in the zone of the PMLF devoid of soot. This process yields $a = 8.1 \text{ s}^{-1}$ and 5.8 s^{-1} for the 1D-CF equivalent to the PMLF cross sections at HAB = 25 and 50 mm, respectively. Additionally, the results of Fig. 2 allow quantifying $Q_{0,0^\circ,soot} = Q_{0,0^\circ f} - Q_{0,0^\circ g}$ which is due to soot particles and appears to be significantly larger at HAB = 50 mm compared to 25 mm, as well as in the seeded compared to the base PMLF. Increasingly larger values of $Q_{0,0^\circ,soot}$ are due to the presence of soot particles characterized by increasingly larger number concentrations and/or sizes.

3.2. Thermochemical structure of the PMLF

The effect of replacing 1500 ppm of C_2H_4 in the fuel stream with the 2,2,4,6,6-pentamethyl-heptane (iC12) isomer of iso-dodecane is presented in Fig. 3 with the profiles of temperature (T), major species, and iC12 in the base and doped PMLFs at HAB = 50 mm and their equivalent 1D-CFs. The trace-level doping has a negligible effect on T , which is directly measured solely in the base PMLF at HAB = 50 mm [25]. However, 1D-CF simulations indicate that T increases by less than 15 K in the doped PMLF compared to the base case, as shown in Table 1. In brief, the comparison of the species in the doped versus the base PMLF is not affected by the small T perturbation and highlights the pyrolysis path of iC12. In Fig. 3 only CH_4 and C_2H_6 increase appreciably (i.e., by 33 % and 36 %, respectively) because of the decomposition of iC12. Also, the 1D-CF mimics well the experimental results, except for the under-prediction of C_2H_6 by 61 % and 83 % in the doped and the base PMLFs, respectively. One can also observe the different slopes and positions of the decay of C_2H_4 and iC12, which is due to different values of

their mass diffusivity. Fig. 4 highlights that the effect of iC12 doping is apparent on all C_3 - C_{18} species with propyne, allene, and benzene being enhanced by 44, 72, and 24 %, respectively. The maximum surge, from 12 to 472 ppm, is observed for the C_4H_8 isomers, prevalently due to iso-butene being cleaved from iC12 via beta-scission [43]. The 1D-CF mimics the surplus of C_3H_4 isomers due to iC12 pyrolysis in the doped PMLF but underpredicts their amounts in the base PMLF. On the other hand, the 1D-CF underestimates twofold the surge in C_4H_8 isomers and overestimates more than two times that of benzene. Fig. 4 also quantifies the noticeable enhancement of most aromatics due to iC12 doping. The 1D-CF mimics the experimental profiles of phenylacetylene and acenaphthylene, but it overpredicts, slightly, phenanthrene and the lumped isomers of naphthalene, and, significantly, cyclopenta[cd]pyrene. It underpredicts slightly styrene and indene, and significantly, the xylene isomers, cyclopenta[def]phenanthrene, and pyrene. Note that mis-predictions are mostly due to flaws in accounting for the chemistry of the base PMLF since the 1D-CF accurately mimics the excess amounts of most aromatics produced as a result of the decomposition of iC12. This observation indicates that the kinetics of iC12 pyrolysis are overall relatively well represented by the LLNL model [17]. A notable exception is benzene, whose concentrations are accurately predicted in the base PMLF but overpredicted in the doped PMLF. As a consequence, the rates of some aromatic growth reactions are likely overpredicted by the LLNL model because the (iC12-doping-caused) concentration enhancement of the larger PAHs formed from benzene growth is overall captured (and in some cases even overpredicted) by the 1D-CF results.

**Fig. 3.** Temperature (T) and mole fraction (X_i) of major species in the base and doped PMLFs at HAB = 50 mm.

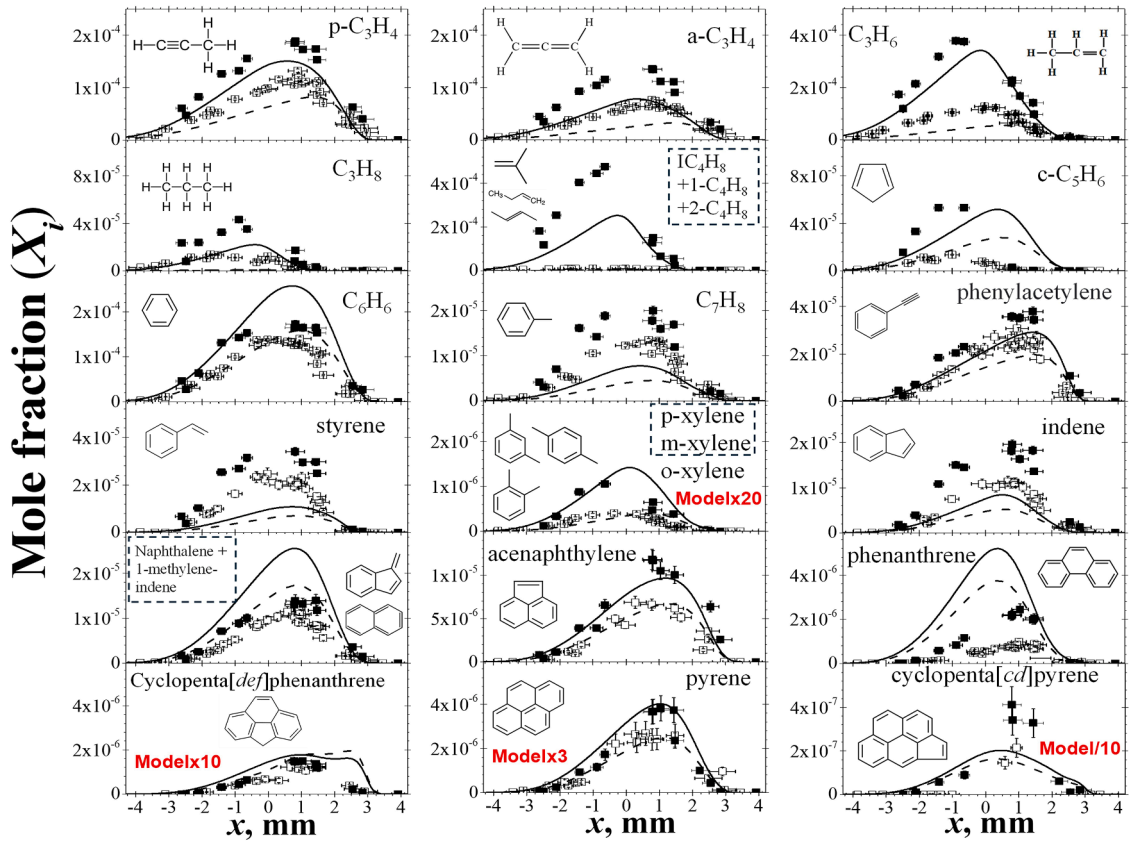


Fig. 4. C_3 - C_{18} species in the base and doped PMLFs at HAB = 50 mm, with legend as in Fig. 3. Species with formulas bracketed in boxes cannot be separated by the performed GC-MS analyses.

3.3. Volume fraction and E-LLS equivalent diameter of soot

Fig. 5 shows the volume fraction (f_v) and E-LLS equivalent diameter ($d_{6,3}$) of soot in the cross-sections at HAB = 25 and 50 mm of the base and doped PMLFs. Symbols and lines identify the results of measurements and the equivalent 1D-CF model, respectively. As expected, since $Z_{st} = 0.183$, the onset of soot inception (i.e., nucleation) occurs in the oxidizer stream right downstream of the blue chemiluminescent flame

front, which tracks T_{max} [25,26,30] and is positioned near $x = 2.7$ and 2.2 mm at HAB = 50 and 25 mm, respectively. Such positioning and the different thicknesses of the profiles mirror the $a = 8.1 \text{ s}^{-1}$ to 5.8 s^{-1} change, involving an approximate doubling of the maximum f_v at HAB = 50 mm compared to that at HAB = 25 mm in both base and doped PMLFs, with the shape of the f_v profiles being approximately the same and peaking at $x \approx 1$ mm. The f_v decrease moving to the left between $x \approx 1$ mm and the GMI is the result of diffusion mixing of the oxidizer into

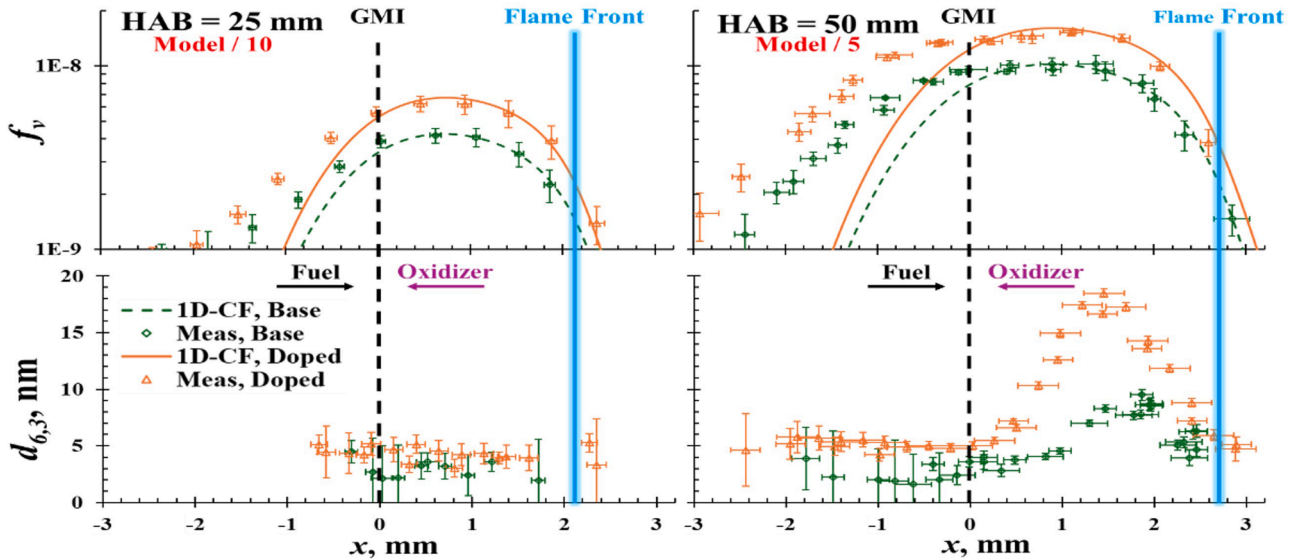


Fig. 5. Soot volume fraction (top panel) and E-LLS equivalent diameter (bottom panel) in the base and doped PMLFs at HAB = 25 (left panel) and 50 mm (right panel), with legend as in Fig. 2. (For interpretation of the references to color in the legend, the reader is referred to the Web version of this article).

the fuel stream, in which soot is less abundant due to nucleation being dampened and, eventually, deactivated as T decreases toward the left. In all cases, $d_{6,3}$ is about 3–5 nm right after nucleation and stays roughly constant at HAB = 25 mm, whereas it grows significantly while the flame products are transported toward the f_v maximum at HAB = 50 mm because of effective coagulation growth [26].

The f_v increase with constant $d_{6,3}$ at HAB = 25 mm indicates nucleation persisting near the GMI, despite low T s, with coagulation not being active [44] during the time longer than 20 ms that the flame products spend in the soot-laden region to reach HAB = 25 mm [25,26]. On the other hand, surface growth may contribute, alongside coagulation, to the $d_{6,3}$ increase in the oxidizer stream at HAB = 50 mm which corresponds to transport times of about 50 ms [25,26].

As for the effect of iC12 doping, Fig. 5 shows that it involves f_v being enhanced by roughly 50 % at both HABs. Such enhancement quantifies the much larger sooting propensity (and carbon content) of iC12 compared to the C_2H_4 it replaces. It is interesting to note that the change in α at different HABs does not affect the f_v enhancement factor caused by doping but only the absolute value of f_v , which is larger at HAB = 50 mm. The results also highlight that iC12 doping increases slightly the sizes of the soot nuclei at HAB = 50 mm because of a substantial boosting of coagulation, due to the f_v increase. Enhanced coagulation also doubles the maximum $d_{6,3}$ from ≈ 9 to ≈ 18 nm. Interestingly, the f_v enhancement is not sufficient to activate soot growth at low T in the fuel stream at HAB = 50 mm, nor anywhere at HAB = 25 mm.

Finally, Fig. 5 assesses also the performance of the LLNL chemical kinetic model [17] in predicting the sooting structure of the PMLF horizontal cross-sections as that of their equivalent 1D-CFs. The shape of the measured f_v profile is mimicked by those in the equivalent 1D-CFs only in the oxidizer stream (i.e., $x > 0$), where the LLNL model overpredicts the f_v absolute values 10 and 5 times at HAB = 25 mm and 50 mm, respectively. It appears that the 1D-CFs account for the high- T soot formation mechanisms but substantially overpredict their nucleation rates, with f_v overpredictions being larger where only nucleation is active (i.e., at HAB = 25 mm compared to 50 mm). Instead, the shape discrepancy of the experimental and 1D-CF profiles to the left of the GMI highlights a low- T nucleation mechanism active in the PMLF [26] which is not accounted for in the LLNL model. When comparing the shape departures of the modeled relative to the experimental profiles, low- T nucleation appears to be more effective at HAB = 50 than at 25 mm. The low- T nucleation becoming more significant at increasing HAB in the PMLF is probably the result of the residence times within it being substantially longer than in other canonical flame configurations in which its effects cannot be appreciated.

4. Conclusions

The pyrolysis and soot formation chemistry of the 2,2,4,6,6-pentamethyl-heptane (iC12) isomer of iso-dodecane has been studied by doping 1500 ppm of it, replacing the same amount of ethylene, in the nitrogen-diluted fuel stream of a PMLF [25]. The key results are summarized below.

- The strain rate of 1D-CFs equivalent to the PMLF horizontal cross-sections is inferred by measuring the profile of their elastic light scattering coefficient.
- The implemented doping approach preserves the PMLF self-similar thermochemical structure in which both iC12 pyrolysis and soot formation occur, a fact that enables focusing on studying their kinetics.
- The pyrolysis of iC12 in the PMLF leads to quantifiable overproductions of many species, as observed with a substantial surge of C_4H_8 and an approximate doubling of a few aromatics.
- The LLNL chemical kinetic model used in a 1D-CF solver with strain rates equivalent to the PMLF horizontal cross-sections mimics the

overall products of iC12 pyrolysis but is quantitatively less accurate in predicting those of the base PMLF.

- The soot volume fraction in the iC12 doped PMLF is 50 % larger than in the base PMLF, probably due to enhanced nucleation of 3–5 nm soot particles occurring both at high Temperature (T) in the oxidizer stream and low T in the fuel stream.
- Coagulation growth is enhanced by iC12 doping in the oxidizer stream only at HAB = 50 mm but is inactive everywhere at HAB = 25 mm and in the fuel stream at HAB = 50 mm.
- The LLNL model accounts for the overall soot formation mechanisms in the oxidizer stream but overpredicts the rate of the high- T nucleation it considers and does not account for the low- T nucleation occurring in the fuel stream.
- A robust experimental methodology and database have been established to improve and validate predictive chemical kinetic models for surrogate blends of traditional Jet Fuels as well as Synthetic Aviation Turbine Fuels (SATFs).

Novelty and significance statement

This work investigates the non-premixed combustion chemistry of 2,2,4,6,6-pentamethyl-heptane, including the effects on soot and its gaseous precursors, by using it as a dopant in the ethylene fuel stream of the novel canonical laminar Planar Mixing Layer Flame (PMLF) configuration. The results of laser diagnostics and capillary sampling for GC-MS analyses track the pyrolysis products of the dopant up to the formation of soot, and they are compared with the predictions of the Lawrence Livermore National Laboratory chemical and soot formation kinetic model. The results are significant because they fill a data gap necessary for validating chemical and soot formation models in conditions of relevance to combustion applications using Synthetic Aviation Turbine (and traditional Jet) Fuels.

CRedit authorship contribution statement

Christian P. Bjork: Investigation - Laser diagnostics, Data curation, Formal analysis, Visualization, Writing - original draft, and Writing - review & editing.

Mahmoud K. Ashour: Investigation - PMLF stabilization, GC/MS measurements, and Computational modeling, Data curation, Formal analysis, Software, Visualization, Writing - original draft, and Writing - review & editing.

Evangelos K. Stefanidis: Investigation - Laser diagnostics, Data curation, Formal analysis.

Chiara Saggese: Conceptualization, Funding Acquisition, Project administration, Investigation - Chemical Kinetic Modeling, and Writing - review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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